

Multi-Layered SOC Lab: Threat Detection & Automated Malware Remediation

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Tech Stack: Wazuh (SIEM), Suricata (IDS), VirusTotal API, Ubuntu (Endpoint), Kali Linux (Manager), Diamorphine.

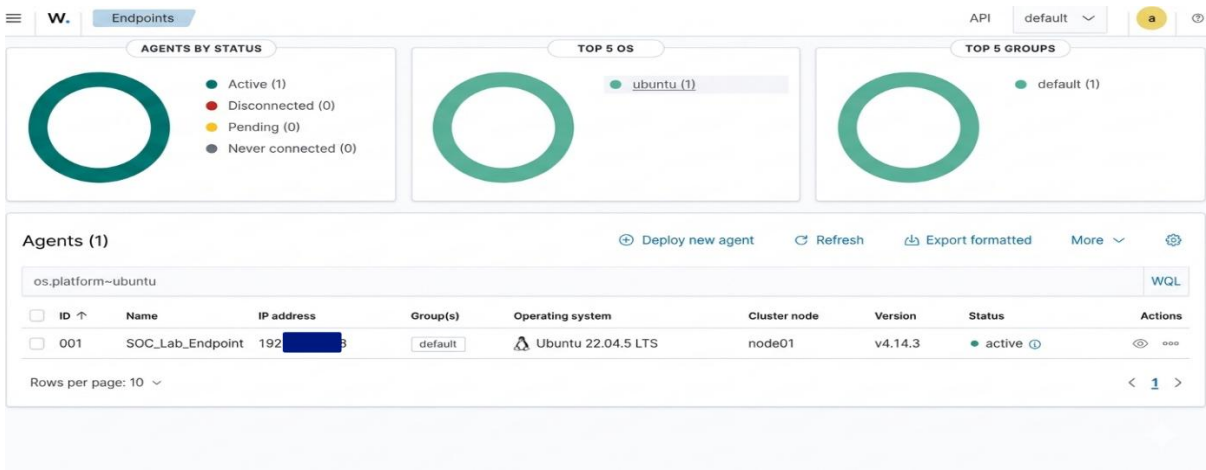
1. Executive Summary

This project demonstrates the deployment of a functional Security Operations Center (SOC) designed to detect host-based, network-based, identity-based and file-based threats. By integrating Wazuh, Suricata, and VirusTotal, I created a unified visibility pipeline that identifies kernel-level obfuscation (rootkits), malicious network patterns, unauthorized access attempts, and automated malware remediation within a virtualized lab environment.

2. Infrastructure Architecture

The lab environment was built using a private virtual network to simulate an enterprise internal LAN.

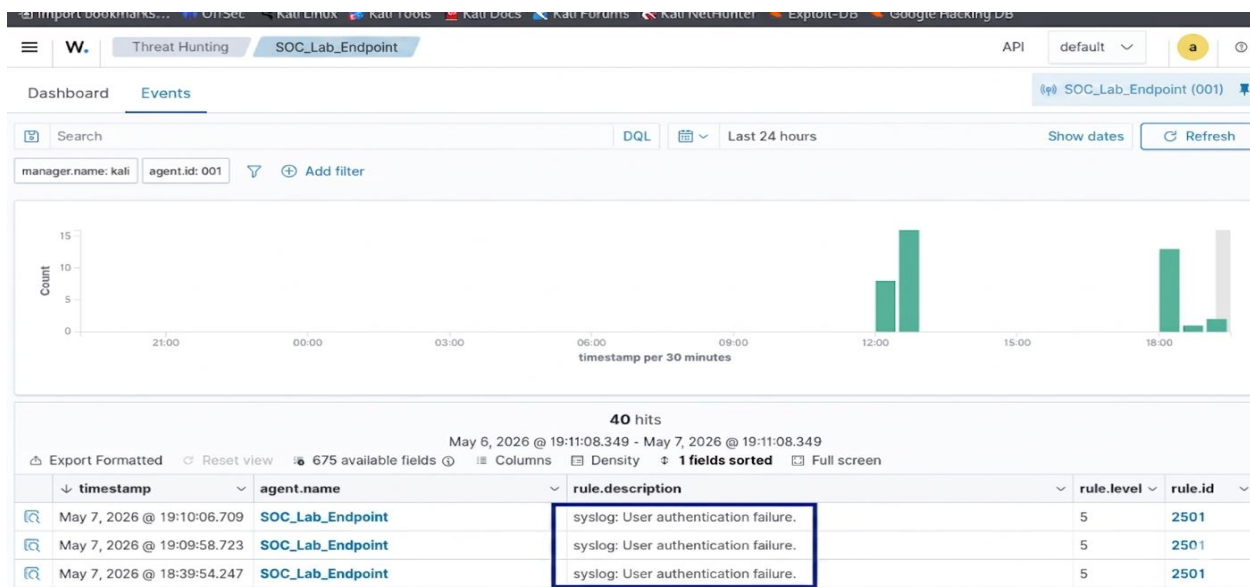
- **Wazuh Manager:** Kali Linux (Centralized Intelligence).
- **Managed Endpoint: Agent Name:** SOC_Lab_Endpoint (Ubuntu 22.04 LTS)
- **Connectivity:** The SOC_Lab_Endpoint agent communicates with the Manager over a dedicated virtual bridge. Connectivity was verified via the Wazuh Dashboard, showing the agent status as "Active."



3. Identity & Access Monitoring: Syslog Analysis

I established baseline visibility for system access.

- **Attack Simulation:** Performed multiple failed login attempts on the SOC_Lab_Endpoint agent.
- **Detection Result:** The Wazuh Dashboard successfully generated "User authentication failure" alerts Rule level 3 and Rule.id 2501. This provided real-time visibility into potential brute-force attacks.



4. Host-Based Security: Rootkit Emulation & Detection

The objective was to test the SIEM's ability to see "invisible" threats using the Diamorphine rootkit.

- **Attack Phase:** Executed a stealth attack to hide the rsyslogd process. Got a PID using **ps aux | head** and run the kill signal Using **kill -31 PID**, the process remained active in memory but became invisible to **ps aux**
- **Defense Phase:** Hardened the SOC_Lab_Endpoint agent by changing the rootcheck frequency to 30 seconds for rapid integrity audits.
- **Detection Result:** Wazuh Manager identified the discrepancy between the kernel's process table and system calls, triggering a Level 11 High Severity Alert (Rule 521) for a hidden process.

```

root@osboxes:/home/osboxes/Diamorphine# ps aux | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 168036 13456 ?        Ss   13:19   0:06 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    13:19   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_gp]
root         4  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_par_gp]
root         6  0.0  0.0      0     0 ?        I<   13:19   0:00 [kworker/0:0H-kblockd]
root         9  0.0  0.0      0     0 ?        I<   13:19   0:00 [mm_percpu_wq]
root        10  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_rude_]
root        11  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_trace]
root        12  2.3  0.0      0     0 ?        S    13:19   3:13 [ksoftirqd/0]
root@osboxes:/home/osboxes/Diamorphine# lsmod | grep diamorphine
kill -63 1
lsmod | grep diamorphine
diamorphine 16384 0
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslogd | grep -v grep
syslog      649  0.0  0.1 222404 4964 ?        Ssl  13:20   0:00 /usr/sbin/rsyslogd -n -iNONE
root@osboxes:/home/osboxes/Diamorphine# kill -31 649
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslog | grep -v grep
root@osboxes:/home/osboxes/Diamorphine#

```



The screenshot shows the Wazuh Threat Hunting interface. At the top, it indicates '640 hits' for the rule 'Possible kernel level rootkit'. The table below lists the hits, showing the timestamp, agent name (SOC_Lab_Endpoint), rule description, rule level (11), and rule ID (521).

timestamp	agent.name	rule.description	rule.level	rule.id
May 7, 2026 @ 20:45:30.773	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.700	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.621	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:29.073	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521

5. Network Security: Suricata IDS Integration

- **Implementation:** Installed Suricata with the Emerging Threats (ET) Open rule set.
- **The Pipeline:** Integrated Suricata's eve.json output into the SOC_Lab_Endpoint Agent's ingestion engine, merging network telemetry with host logs.
- **Validation:** Generated ICMP traffic via the Kali Linux terminal. The IDS successfully flagged the reconnaissance attempt, providing a unified view on the Wazuh Dashboard.

```
osboxes@osboxes: /tmp
GNU nano 6.2 /var/ossec/etc/ossec.conf
<active-response>
  <disabled>no</disabled>
  <ca_store>etc/wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active-response>

<!-- Choose between "plain", "json", or "plain,json" for the format of intern
<logging>
  <log_format>plain</log_format>
</logging>

</ossec_conf>

<ossec_config>
  <localfile>
    <log_format>journald</log_format>
    <location>journald</location>
  </localfile>

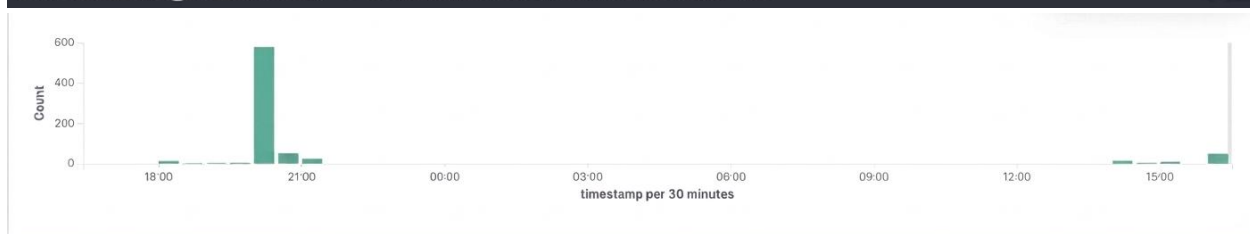
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>
</ossec_config>
```

```
Session Actions Edit View Help
kali)-[~]
$ ping -c 20 192.168.1.8
PING 192.168.1.8 (192.168.1.8) 56(84) bytes of data.
64 bytes from 192.168.1.8: icmp_seq=1 ttl=64 time=1.06 ms
64 bytes from 192.168.1.8: icmp_seq=2 ttl=64 time=0.898 ms
64 bytes from 192.168.1.8: icmp_seq=3 ttl=64 time=1.43 ms
64 bytes from 192.168.1.8: icmp_seq=4 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=5 ttl=64 time=1.10 ms
64 bytes from 192.168.1.8: icmp_seq=6 ttl=64 time=1.35 ms
64 bytes from 192.168.1.8: icmp_seq=7 ttl=64 time=0.977 ms
64 bytes from 192.168.1.8: icmp_seq=8 ttl=64 time=1.77 ms
64 bytes from 192.168.1.8: icmp_seq=9 ttl=64 time=1.37 ms
64 bytes from 192.168.1.8: icmp_seq=10 ttl=64 time=0.923 ms
64 bytes from 192.168.1.8: icmp_seq=11 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.1.8: icmp_seq=13 ttl=64 time=1.31 ms
64 bytes from 192.168.1.8: icmp_seq=14 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=15 ttl=64 time=1.46 ms
64 bytes from 192.168.1.8: icmp_seq=16 ttl=64 time=1.18 ms
64 bytes from 192.168.1.8: icmp_seq=17 ttl=64 time=1.36 ms
64 bytes from 192.168.1.8: icmp_seq=18 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=19 ttl=64 time=1.17 ms
64 bytes from 192.168.1.8: icmp_seq=20 ttl=64 time=0.955 ms

— 192.168.1.8 ping statistics —
20 packets transmitted, 20 received, 0% packet loss, time 19226ms
rtt min/avg/max/mdev = 0.898/1.232/1.771/0.218 ms
```



750 hits

May 7, 2026 @ 16:24:48.281 - May 8, 2026 @ 16:24:48.281

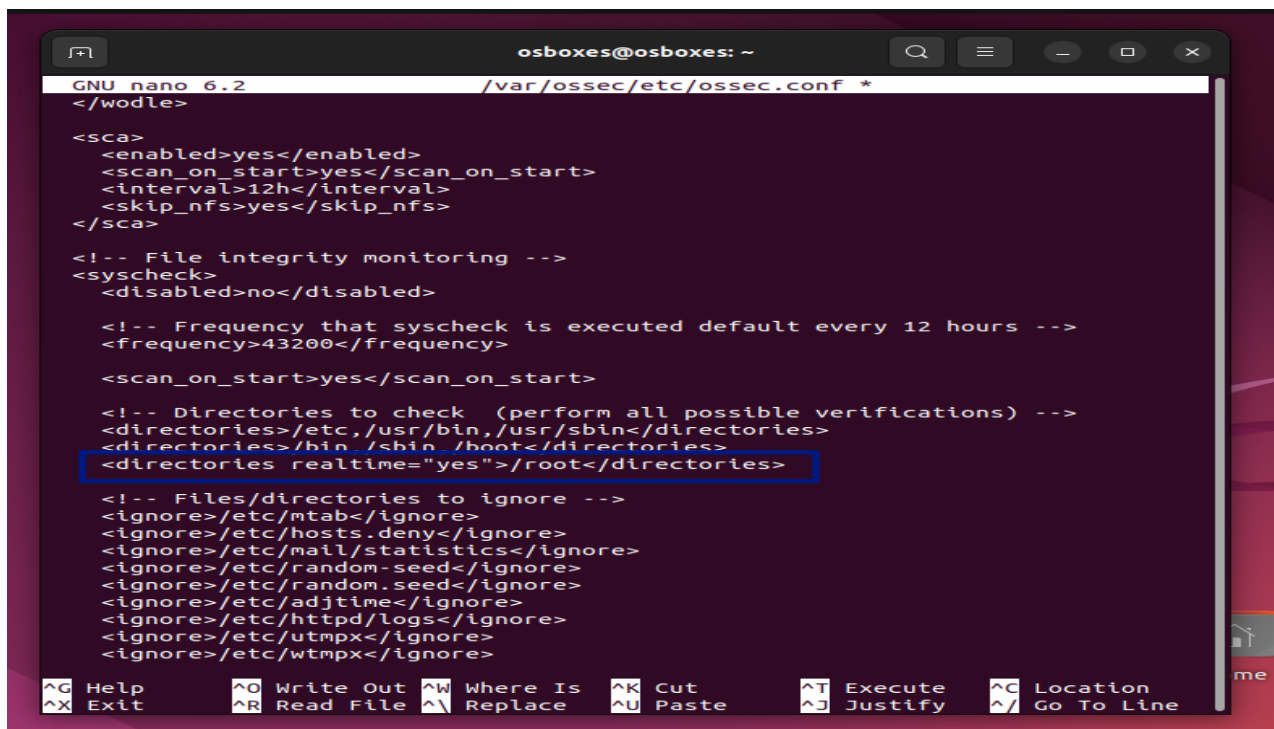
Export Formatted Reset view 684 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
May 8, 2026 @ 16:22:08.468	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:08.466	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:06.465	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:06.462	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601

6. Automated Response: VirusTotal & Malware Remediation

Implemented an **Active Response** loop to handle malicious files automatically.

- **File Integrity Monitoring (FIM):** Configured SOC_Lab_Endpoint agent to monitor the /root directory for new files.
- **API Integration:** Connected Wazuh Manager to the VirusTotal API to analyze file hashes detected on the SOC_Lab_Endpoint agent via FIM.
- **Threat Trigger:** Created a custom rule (**87105**) that fires only when VirusTotal returns a positive malware result.
- **Validation:** I downloaded the **EICAR** test file to the SOC_Lab_Endpoint /root directory. Within seconds, the file was identified, verified as malicious, and deleted.



```
osboxes@osboxes: ~  
GNU nano 6.2 /var/ossec/etc/ossec.conf *  
</wodle>  
  
<sca>  
  <enabled>yes</enabled>  
  <scan_on_start>yes</scan_on_start>  
  <interval>12h</interval>  
  <skip_nfs>yes</skip_nfs>  
</sca>  
  
<!-- File integrity monitoring -->  
<syscheck>  
  <disabled>no</disabled>  
  
  <!-- Frequency that syscheck is executed default every 12 hours -->  
  <frequency>43200</frequency>  
  
  <scan_on_start>yes</scan_on_start>  
  
  <!-- Directories to check (perform all possible verifications) -->  
  <directories>/etc,/usr/bin,/usr/sbin</directories>  
  <directories>/bin /sbin /boot</directories>  
  <directories realtime="yes">/root</directories>  
  
  <!-- Files/directories to ignore -->  
  <ignore>/etc/mtab</ignore>  
  <ignore>/etc/hosts.deny</ignore>  
  <ignore>/etc/mail/statistics</ignore>  
  <ignore>/etc/random-seed</ignore>  
  <ignore>/etc/random.seed</ignore>  
  <ignore>/etc/adjtime</ignore>  
  <ignore>/etc/httpd/logs</ignore>  
  <ignore>/etc/utmpx</ignore>  
  <ignore>/etc/wtmpx</ignore>  
</syscheck>  
</ossec>
```



```
Session Actions Edit View Help
GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<skip_nfs>yes</skip_nfs>

<ignore>/var/lib/containerd</ignore>
<ignore>/var/lib/docker/overlay2</ignore>
</rootcheck>

<wodle name="cis-cat">
  <disabled>yes</disabled>
  <timeout>1800</timeout>
  <interval>id</interval>
  <scan-on-start>yes</scan-on-start>

  <java_path>wodles/java</java_path>
  <ciscat_path>wodles/ciscat</ciscat_path>
</wodle>

<!-- Osquery integration -->
<wodle name="osquery">
  <disabled>yes</disabled>
  <run_daemon>yes</run_daemon>
  <log_path>/var/log/osquery/osqueryd.results.log</log_path>
  <config_path>/etc/osquery/osquery.conf</config_path>
  <add_labels>yes</add_labels>
</wodle>

<integration>
  <name>virstotal</name>
  <api_key>09b42c5b5afc29c35ea680057a</api_key>
  <rule_id>100200,100201</rule_id>
  <alert_format>json</alert_format>
</integration>

<!-- System inventory -->
<wodle name="syscollector">
```

```
Session Actions Edit View Help
GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<executable>netsh.exe</executable>
<timeout_allowed>yes</timeout_allowed>
</command>

<command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
</command>

<!--
<active-response>
  active-response options here
</active-response>
-->

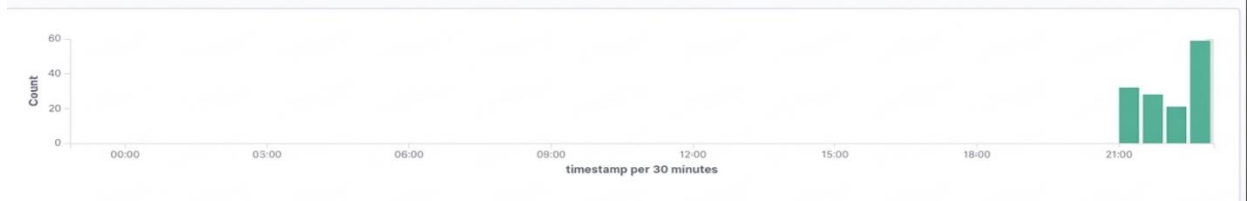
<active-response>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>

<!-- Log analysis -->
<localfile>
  <log_format>command</log_format>
  <command>df -P</command>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>full_command</log_format>
  <command>netstat -tulpn | sed 's/([[:alnum:]]+)[[:space:]]+([[:digit:]]+)[[:space:]]+([[:digit:]]+)[[:space:]]+([[:digit:]]+)</command>
  <alias>netstat listening ports</alias>
  <frequency>360</frequency>
</localfile>
```

Search [DQL] [Last 24 hours] [Show dates] [Refresh]

manager.name: kali agent.id: 001 [Add filter]



140 hits

May 9, 2026 @ 22:50:13.058 - May 10, 2026 @ 22:50:13.058

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timestamp	agent.name	rule.description	rule.level	rule.id
May 10, 2026 @ 22:49:47.8...	SOC_Lab_Endpoint	File deleted.	7	553
May 10, 2026 @ 22:49:47.7...	SOC_Lab_Endpoint	VirusTotal: Alert - /root/eicar.com - 66 engines detected this file	12	87105
May 10, 2026 @ 22:49:38.9...	SOC_Lab_Endpoint	PAM: Login session closed.	3	5502
May 10, 2026 @ 22:49:37.3...	SOC_Lab_Endpoint	File added to /root	7	100201
May 10, 2026 @ 22:49:36.9	SOC_Lab_Endpoint	PAM: Login session opened	3	5501

```
osboxes@osboxes: ~  
osboxes@osboxes:~$ sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
                                 Dload  Upload   Total   Spent    Left   Speed  
100    68  100    68    0     0    71      0  --:--:-- --:--:-- --:--:--    71  
osboxes@osboxes:~$ sudo ls -la /root/eicar.com  
ls: cannot access '/root/eicar.com': No such file or directory  
osboxes@osboxes:~$
```

7. Conclusion

This lab validates a defense-in-depth strategy. While the rootkit (Diamorphine) successfully fooled local OS utilities and network probes (ICMP) tested the perimeter, the SOC's ability to cross-reference kernel data, inspect network packets, and autonomously remediate malware through API integrations ensures that even sophisticated attacks are identified and neutralized.